When a High Score Is an Illusion: Certifying Genuine versus Repackaged Forecasting Skill

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***Abstract*—A forecast that repeats last week's sales can still earn a high score. We ask whether it contains information beyond a declared baseline and whether evaluation itself creates the apparent association. Our contribution separates rank-reference reuse, imperfect baseline adjustment and temporal feedback. For independently sampled peers, using distinct references removes reuse bias, including ties. With exact category masses, split validation bounds the aggregate learning error directly, avoiding simultaneous intervals for each conditional mean. Subtracting this allowance and classical sampling uncertainty gives a finite-sample certificate under independent reference sampling. For forecasts built from past outcomes, a second identity identifies which training folds create feedback and when earlier/later fitting removes it in the stated raw-score model. With learned controls in a matched null simulation, distinct references reduce rejections from 402 to 41 out of $1{,}000$ panels. In a simulation with 32 balanced categories, $8{,}192$ validation pairs and exact mean fits, aggregate validation reduces median bias allowance from 0.018517 to 0.002602. Analyses of 41 public predictors measure evaluation sensitivity; forecast studies separately measure augmentation gains and certification value.**

***Index Terms*—forecast evaluation, residual covariance, bias correction, rank statistics, multiple testing, panel forecasting**

I. Introduction

A demand forecast can rank well by repeating last week's sales. To find out whether it adds information, an analyst predicts both the forecast and next week's sales from this baseline, then tests whether the unexplained parts move together. This residual audit builds on forecast encompassing [1] and residual-covariance tests [2].

Evaluation can create the association it is supposed to measure. Ranking forecast and sales against the same products reuses random references. Imperfect baseline adjustment leaves a separate error. Training on later forecasts can also feed the evaluated week's sales back into its fitted controls. More observations do not automatically remove these effects.

a) Preserve the target before testing: Exact identities separate the reference and temporal mechanisms. Distinct references remove reuse bias for independent peer samples, including ties. Remaining bias is the product of two fitting errors. Independent validation bounds it with its sign; known category masses permit a direct aggregate bound. The certificate subtracts learning and sampling allowances from corrected association. Declared controls define what "beyond" means: high/low sales categories leave within-category variation in last week's sales. Certification concerns that stated representation under the reference design.

b) Choose the audit from the available information: Table I maps information to actions and conclusions. Matched experiments isolate correction, learning allowances and classical sampling bounds. Four tasks keep 41 forecasts fixed while varying the audit; separate studies measure augmentation gains and certification value.

II. Related Work

a) What residual evidence means: Forecast encompassing [1] concerns information beyond another forecast; conditional predictive-ability tests compare losses [3]. Our positive residual-covariance target differs from equal accuracy and arbitrary conditional dependence. The generalized covariance measure (GCM) [2] and cross-fitting [4], [5] provide the orthogonal-score argument. Kernel [6], partial-copula [7], projected-covariance [8] and weighted-GCM [9] tests address related targets; recent studies examine practical calibration [10], [11].

b) What the corrections add: Leading-bias cancellation [12], coarsening corrections [13] and local permutation [14] require their own approximation conditions. Our contribution specifies reference reuse and temporal-feedback accounting for the audit. U-statistic concentration [15], [16] and bounded-mean betting [17] supply classical sampling comparators. Known forecast metadata connect to Gini covariance [18] and semi-supervised U-statistics [19]. Bias falsification [20] tests rate assumptions; our finite-category bounds use independent validation. Direct aggregate bias estimation is classical [21]; our result specifies a finite rank-target allowance under exact category metadata.

c) Why temporal fitting matters: Dynamic fixed-effects bias [22], recursive mean adjustment [23], forward orthogonal deviations [24] and separated-block inference [25] provide the temporal background. We derive a fold-specific decomposition including sign reversal and a raw-score specialization. The classical rank-correlation decomposition [26] is attributed; our reference theorem connects reuse and estimated-mean errors to the group-size ratio.

III. Methodology

A. Target and interpretation

In the demand example, X is the forecast, Y is realized sales and Z includes the declared representation of last week's sales. The same notation applies on any stated scale; panel

TABLE I

CHOOSE THE ROW BEFORE EVALUATION. RESIDUAL CERTIFICATES CONCERN DECLARED CONTROLS; PREDICTION GAINS REQUIRE SEPARATE LOSS COMPARISONS.

Available information and sampling	Audit to use	What the result supports
Independent peers. Common law; estimated rank distributions; independent training; declared categories	Remove reference reuse. With exact category masses, validate aggregate bias; otherwise use mean rectangles. Apply a fixed sampling bound.	A finite lower bound on midrank association ϑ; asymptotic tests need the stated learning rates and variance.
Known forecast ranks. Complete metadata; independent archive draws	Compare known-forecast covariance kernels or pairs. Complete labeled archives also permit exact census.	Association within categories without learning both means; account for the extra metadata.
Fixed peers. Conditionally independent channels and entities; exact means	Use shared references under Eq. (13).	A centered fixed-roster score; distinct references can introduce bias.
Temporal forecasts. Predictable raw scores; independent entities under Proposition 3	Fit forecast effects earlier and outcome effects later; use entity variance.	Inference for a raw directional moment, not necessarily θ_∞ under alternatives.
Observed rank panels. Additive controls and declared folds	Compare prespecified bin, spline and fitting choices; retain all raw results.	Sensitivity evidence; confirm an inferential rule separately before declaring discoveries.

TABLE II

NOTATION: KEEP THE TARGET SEPARATE FROM ITS ERROR ALLOWANCES.

Symbol	Meaning
θ_∞	Fully adjusted covariance on the declared scale
ϑ	Marginal-midrank residual covariance
$\theta_{\mathcal{A}}$	Covariance after additive-class adjustment
$M, N;\ G$	Groups and peers per group; entity trajectories
$b_H;\ \widehat{B}$	Learning bias; its selected upper allowance
$\bar{D}, r_U, L_\alpha$	Corrected mean, sampling allowance, lower bound

analyses use standardized within-period average ranks. Let $m_X = \mathbb{E}[X \mid Z]$, $m_Y = \mathbb{E}[Y \mid Z]$ and $r_X = X - m_X$, $r_Y = Y - m_Y$. We test

$$\theta_\infty = \mathbb{E}[r_X r_Y], \qquad H_0 : \theta_\infty \le 0 \quad \text{against} \quad H_1 : \theta_\infty > 0. \tag{1}$$

If $v_X = \mathbb{E}r_X^2 > 0$, the best augmentation $m_Y + t r_X$, $t \ge 0$, reduces squared-error risk by $(\theta_\infty)_+^2 / v_X$: expand its risk as $\mathbb{E}r_Y^2 - 2t\theta_\infty + t^2 v_X$. For any fitted $\widehat{t}$ and positive θ_∞, the gain is $\theta_\infty^2 / v_X - v_X(\widehat{t} - \theta_\infty / v_X)^2$. Certification concerns positive residual association under the stated sampling model. Zero covariance permits nonlinear dependence.

For a_X, a_Y that are population fits in a declared additive class, its residual target is

$$\begin{aligned} \theta_{\mathcal{A}} &:= \mathbb{E}[(X - a_X)(Y - a_Y)] \\ &= \theta_\infty + \mathbb{E}[(m_X - a_X)(m_Y - a_Y)]. \end{aligned} \tag{2}$$

Conditional centering removes both cross terms. Interpreting additive panel fits as θ_∞ requires the second term to be negligible. Ranking also changes the target: raw conditional independence need not survive conditioning on observed baseline ranks.

B. Rank references and learning error

To rank a product, the sales audit compares it with reference products. Consider M independent groups of $N \ge 3$ i.i.d. raw triples (V_i, W_i, Z_i) from a common law, with nuisance fits $f, g \in [0, 1]$ fixed by independent training H. Let $a(v, v') = \mathbf{1}\{v' < v\} + \frac{1}{2}\mathbf{1}\{v' = v\}$, $a_{ij} = a(V_i, V_j)$, and $b_{ij} = a(W_i, W_j)$. Set $k = N - 1$, $U_{V,i} = k^{-1} \sum_{j \ne i} a_{ij}$ and $U_{W,i} = k^{-1} \sum_{j \ne i} b_{ij}$. Let $F_V(v) = \Pr(V < v) + \frac{1}{2}\Pr(V = v)$ and define F_W analogously. Write $R_V = F_V(V)$, $R_W = F_W(W)$, $m_V = \mathbb{E}[R_V \mid Z, H]$ and m_W analogously. The target is $\vartheta = \mathbb{E}[(R_V - m_V)(R_W - m_W) \mid H]$, separate from standardized panel ranks. The shared-reference score $C_i = (U_{V,i} - f(Z_i))(U_{W,i} - g(Z_i))$ becomes

$$D_i = C_i + \frac{U_{V,i} U_{W,i} - k^{-1} \sum_{j \ne i} a_{ij} b_{ij}}{k - 1}. \tag{3}$$

This uses distinct reference indices without discarding observations. Given fitted means, the correction needs only two marginal comparison sums and one joint-product sum per row. Sorting and cumulative counts evaluate all N scores in $O(N \log N)$ time and $O(N)$ working memory, including ties.

Proposition 1 (Reference-reuse decomposition)**.** *Write $b_H = \mathbb{E}[(m_V - f)(m_W - g) \mid H]$ and $\Gamma = \mathbb{E}[a_{12} b_{12} - F_V(V_1) F_W(W_1) \mid H]$. Then, including ties,*

$$\mathbb{E}[C_i \mid H] = \vartheta + b_H + \Gamma / k, \qquad \mathbb{E}[D_i \mid H] = \vartheta + b_H. \tag{4}$$

Distinct references remove reuse bias exactly; either exact nuisance removes b_H. The Cauchy-Schwarz inequality gives $|b_H| \le \varepsilon_f \varepsilon_g$, where $\varepsilon_f^2 = \mathbb{E}[(m_V - f)^2 \mid H]$ and similarly for g. Thus the correction removes Γ / k even with imperfect fits; the remaining fitting error depends on their product, not their sum.

Set $r_V = R_V - m_V$, $r_W = R_W - m_W$ and, for an independent copy O',

$$\begin{aligned} h_V(v) &= \mathbb{E}[(a(V', v) - m_V(Z')) r'_W], \\ h_W(w) &= \mathbb{E}[r'_V (a(W', w) - m_W(Z'))], \\ \psi &= r_V r_W + h_V(V) + h_W(W) - 3\vartheta, \quad \sigma_\psi^2 = \mathbb{E}\psi^2. \end{aligned} \tag{5}$$

These terms account for an observation's focal and two reference roles.

Theorem 1 (Covariance boundary and reference reuse)**.** *Let $M, N \to \infty$ under a fixed common law, with exact means $f = m_V$, $g = m_W$ and $\sigma_\psi^2 > 0$. Let A_j be group j's mean score for*

$A=C,D$, $\bar{A}=M^{-1}\sum_j A_j$, $\widehat{v}_A=(M-1)^{-1}\sum_j(A_j-\bar{A})^2$ *and* $T_A=\sqrt{M}\bar{A}/\sqrt{\widehat{v}_A}$*, set to zero at zero variance. Then*

$$\begin{aligned}\sqrt{MN}(\bar{D}-\vartheta)&\Rightarrow\mathcal{N}(0,\sigma_\psi^2),\\ \sqrt{MN}\{\bar{C}-\vartheta-\Gamma/(N-1)\}&\Rightarrow\mathcal{N}(0,\sigma_\psi^2),\\ N\widehat{v}_A&\to_p\sigma_\psi^2.\end{aligned}\tag{6}$$

At every such boundary law $\vartheta=0$, $T_D\Rightarrow\mathcal{N}(0,1)$. *At this boundary, if* $M/N\to\lambda<\infty$*, then* $T_C\Rightarrow\mathcal{N}(\Gamma\sqrt{\lambda}/\sigma_\psi,1)$*; if* $\Gamma>0$ *and* $M/N\to\infty$*, its one-sided rejection probability tends to one. At fixed negative or positive* ϑ*, the corrected rejection probability tends to zero or one, respectively. Under* $V\perp W\mid Z$, $h_V=h_W=0$ *and* $\sigma_\psi^2=\mathbb{E}[r_V^2r_W^2]$.

The group variance estimates all three roles without fitting h_V,h_W. Thus zero covariance need not mean conditional independence. At fixed N, positive group variance gives the ordinary group CLT. For common-law arrays with $\sigma^2_{\psi,M,N}\to\sigma^2_\psi>0$ and $\sqrt{MN}\,\vartheta_{M,N}\to d$, the same argument gives $T_D\Rightarrow\mathcal{N}(d/\sigma_\psi,1)$.

Corollary 1 (Estimated conditional means)**.** *Replace the oracle means in Theorem 1 by fits* $f,g\in[0,1]$ *trained on a common sample independent of all evaluation groups. If* $\varepsilon_f+\varepsilon_g=o_p(1)$ *and* $\sqrt{MN}\,\varepsilon_f\varepsilon_g=o_p(1)$*, all its studentized limits remain valid.*

a) Learning and sampling allowances: For C_Z categories, validation pairs (O,O') are independent of training and evaluation. Focal Z,V,W and reference V',W' supply comparisons with conditional means $m_V(c),m_W(c)$. With n_c focal rows, their Hoeffding intervals have radius $\sqrt{\log(8C_Z/\delta)/(2n_c)}$ and are intersected with $[0,1]$; empty cells receive $[0,1]$. Let q_c^+,q_c^- be the largest and smallest of the four endpoint products $(x-f(c))(y-g(c))$ over these two intervals. Obtain binomial upper limits u_c for category probabilities, each with error $\delta/(2C_Z)$ [27]. Define

$$\widehat{B}_+=\max_{p\in\mathcal{P}}\sum_c p_cq_c^+,\qquad \widehat{B}_-=\min_{p\in\mathcal{P}}\sum_c p_cq_c^-.\tag{7}$$

Here $\mathcal{P}=\{p\geq0:\mathbf{1}^Tp=1,\ p\leq u\}$. Sorted coefficient allocation gives the extrema, including negative terms. With probability at least $1-\delta$, $\widehat{B}_-\leq b_H\leq\widehat{B}_+$. The signed upper allowance can be negative and is no larger than $\widehat{B}_{\rm abs}$ from the largest absolute endpoint errors.

b) Validate the aggregate when category masses are known: The learning bias is one weighted sum. Suppose exact category probabilities p_c are known from control metadata or design; complete forecast or outcome rank metadata are not required. Split independent validation pairs into two streams before inspecting values. In category c, average $a(V,V')-f(c)$ in stream A and $a(W,W')-g(c)$ in stream B, giving $\widehat{a}_c,\widehat{b}_c$ and counts n_{Ac},n_{Bc}. The pairs still cost two raw observations each; splitting adds no queries.

Lemma 1 (Aggregate validation allowance)**.** *Condition on training and all focal validation category assignments. For* $\mathcal{S}=\{c:n_{Ac}n_{Bc}>0\}$*, let* $x=\log(3/\delta)$, $d_c=p_c/(4\sqrt{n_{Ac}n_{Bc}})$ *and*

$$\begin{aligned}r_A^2&=\frac{x}{2}\sum_{c\in\mathcal{S}}\frac{p_c^2\widehat{b}_c^2}{n_{Ac}}, & r_B^2&=\frac{x}{2}\sum_{c\in\mathcal{S}}\frac{p_c^2\widehat{a}_c^2}{n_{Bc}},\\ r_2&=\sqrt{2x\sum_{c\in\mathcal{S}}d_c^2}+x\max_{c\in\mathcal{S}}d_c.\end{aligned}\tag{8}$$

Take empty sums and maxima as zero. Let $q_c^{\rm full}$ *maximize* $(u-f(c))(v-g(c))$ *over* $u,v\in\{0,1\}$. *Then*

$$\widehat{B}_{\rm agg}=\sum_{c\in\mathcal{S}}p_c\widehat{a}_c\widehat{b}_c+r_A+r_B+r_2+\sum_{c\notin\mathcal{S}}p_cq_c^{\rm full}\tag{9}$$

satisfies $\Pr(b_H>\widehat{B}_{\rm agg})\leq\delta$*, including ties.*

This is a finite rank-target specialization of classical second-order bias estimation [20], [21] and concentration. It replaces simultaneous mean rectangles by one observable aggregate bound. Exact category masses are essential; plugging in empirical frequencies requires another uncertainty argument.

For exact fits and $n/C\geq1$ pairs per category per stream with $p_c=1/C$,

$$\mathbb{E}[r_A+r_B+r_2]\leq\frac{3\sqrt{Cx/8}+x/4}{n}.\tag{10}$$

where expectation conditions on these balanced counts.

Proposition 2 (Observable reference certificate)**.** *Under the independent common-law reference design, let* $\widehat{B}$ *be an independently validated upper bound on* b_H *with failure probability* δ*. Let* R *be the range of* $(a-f(c))(b-g(c))$ *over* $a,b\in[0,1]$ *and all categories. Fix disjoint triples before inspecting their values. Let* s^2 *be the sample variance of their six-role symmetric kernels, with denominator* $J-1$*, where* $J=M\lfloor N/3\rfloor\geq2$*. For* $0<\delta<\alpha<1$*, write* $x=\log\{2/(\alpha-\delta)\}$ *and*

$$\begin{aligned}c_J&=\frac{2R}{\sqrt{J(J-1)}}+\frac{R}{3J},\\ r_U(\alpha)&=\min\left\{R\sqrt{\frac{x}{2J}},\,s\sqrt{\frac{2x}{J}}+c_Jx\right\},\end{aligned}\tag{11}$$

$$L_\alpha=\bar{D}-\widehat{B}-r_U(\alpha).\tag{12}$$

Then $\Pr(L_\alpha>\vartheta)\leq\alpha$*, including ties and zero kernel variance. The mean* $\bar{D}$ *uses all distinct triples; only its variance estimate uses the fixed disjoint triples.*

This combines classical U-statistic concentration and variance bounds [15], [16] with either preselected allowance. Use $\widehat{B}=\widehat{B}_+$ or $\widehat{B}_{\rm agg}$; taking their random minimum requires a joint error budget. Its inverted $p_{\rm cert}$ is superuniform under $\vartheta\leq0$. A range-only alternative replaces r_U by $R\sqrt{\log\{1/(\alpha-\delta)\}/(2J)}$ and also allows $J=1$. Choose the construction before evaluation. For BY at level α, $\delta=\rho\alpha/(KH_K)$, $0<\rho<1$, places the p-value floor below the first family threshold; $H_K=\sum_{j=1}^K1/j$.

Count training, both validation-pair members and evaluation. The companion retains the rectangular power condition;

$$\begin{tikzcd}
\text{Earlier folds } E & \text{Evaluation } r \in I & \text{Later folds } L \\
\bar{x}_E \arrow[r] & x_r - \bar{x}_E & \\
& y_r - \bar{y}_L & \bar{y}_L \arrow[l]
\end{tikzcd}$$

Fig. 1. **Directional fitting for an audit.** Fit the score mean earlier and the outcome mean later, then multiply the two residuals. Evaluation rows require both sides. Forecast training is unchanged.

mixing its lower allowance with an aggregate upper bound requires a joint error budget.

For two equally likely categories with midrank means $(0.35, 0.65)$ in both channels, fits $f = (0.5, 0.8)$ and $g = (0.2, 0.5)$ give $b_H = -0.0225$. At $\vartheta = 0.02$, absolute subtraction approaches -0.025 as validation and evaluation grow, whereas signed subtraction approaches 0.02. Rank construction fixes the marginal mean $1/2$, not these category means.

c) Which peer population is being audited?: The reference correction depends on the sampling design. Condition on a fixed set of N entity attributes, assume that entity pairs are independent across entities and that the two channels are independent within every entity, and let $f_i = \mathbb{E}[U_{V,i}]$ and $g_i = \mathbb{E}[U_{W,i}]$ be the exact entity means. Then

$$\mathbb{E}[C_i] = 0, \qquad \mathbb{E}[D_i] = -\frac{\mathrm{Cov}_{j \neq i}(p_{ij}, q_{ij})}{k-1}, \tag{13}$$

where $p_{ij} = \mathbb{E}[a_{ij}]$, $q_{ij} = \mathbb{E}[b_{ij}]$, and the covariance weights each of the k peers equally. Distinct references therefore remove a sampling bias for i.i.d. resampled peers but can introduce one for a fixed heterogeneous peer set. Peer recurrence and persistent entity effects are reported separately; neither identifies a valid null model from the observations alone.

C. Feedback through entity effects

A forecast can use an entity's earlier outcomes even when its next outcome is unpredictable. Fitting both entity effects on all other folds then creates two feedback routes: earlier training outcomes enter the evaluated forecast, and the evaluated outcome enters later training forecasts. The following calculation separates these routes from the product of the fitting errors.

Proposition 3 (Feedback under temporal splitting)**.** *For one entity, let $y_r = \alpha + \varepsilon_r$ and $x_r = \sum_{j<r} w_{rj} y_j + \eta_r$, with deterministic weights and contiguous time folds. The square-integrable shocks have mean zero, variance $\sigma^2 > 0$, and are mutually uncorrelated and uncorrelated with α and every η_s. For an evaluation fold I, let E, L be its strictly earlier and later rows, $O = E \cup L$, and $m = |O| > 0$. Write $w_{sj} = 0$ for $j \geq s$. With both entity means fitted on O, set $S_I = \sum_{r \in I} (x_r - \bar{x}_O)(y_r - \bar{y}_O)$. Then*

$$\frac{\mathbb{E} S_I}{\sigma^2} = -\frac{\sum_{r \in I, j \in E} w_{rj}}{m} - \frac{\sum_{s \in L, r \in I} w_{sr}}{m} + \frac{|I|}{m^2} \sum_{s,j \in O} w_{sj}. \tag{14}$$

If E, L are nonempty, then $\mathbb{E}[(x_r - \bar{x}_E)(y_r - \bar{y}_L)] = 0$ for every $r \in I$. This latter result also holds for any square-integrable forecast measurable from α, the forecast noises and outcomes before r, provided $\mathbb{E}[\varepsilon_s \mid \alpha, \eta_1, \ldots, \eta_n, \varepsilon_1, \ldots, \varepsilon_{s-1}] = 0$ for every s.

The first two terms oppose nonnegative forecast weights; the third can reverse their sign. For K folds of $h \geq p$ rows and a forecast equal to a times the sum of the previous p available outcomes, the total expected numerator is $a\sigma^2 K p(h-p-1)/\{(K-1)h\}$. The temporal split therefore fits the forecast effect on earlier folds and the outcome effect on later folds. It adapts recursive mean adjustment and forward orthogonal deviations [23], [24] to the two nuisance fits. Proposition 3 establishes centering for raw outcomes with entity means; the rank and baseline-adjusted panel procedure is evaluated separately in the experiments.

Corollary 2 (Independent-entity inference)**.** *Let S_g be entity g's mean directional product over fixed eligible rows with nonempty earlier and later folds. Suppose entities, including their forecast noises, are independent and satisfy Proposition 3's centering conditions. If $\sup_g \mathbb{E}|S_g|^4 < \infty$ and $G^{-1} \sum_{g=1}^{G} \mathrm{Var}(S_g) \to v > 0$, then $\sqrt{G}\, \bar{S} / \widehat{\mathrm{sd}}(S) \Rightarrow \mathcal{N}(0,1)$.*

This equal-entity raw-score test allows fixed observation counts per entity. It does not require consistent estimation of each entity effect; its centering relies on the directional design. It is distinct from the pooled-row ranked panel screen.

D. Observed-panel sensitivity and selection

Panel screens use cross-fitted additive entity effects and baseline bins, leaving products $R_i^{(q)} = e_{X,i}^{(q)} e_{Y,i}^{(q)}$ at resolution q. Ordered panels use Newey–West variance at the stated Bartlett lag [28]; ratings use cluster variance. The companion gives both formulas. A cluster CLT, consistent variance and negligible fitted product error are needed for valid inference [2], [5]; these choices alone do not establish them.

An outcome-free rule abstains on exact copies, matching or reversed cluster orders with ties, and constant ranks. Setting $p^* = 1$ preserves valid inputs because $p^* \geq p$, without repairing invalid ones. For resolution sensitivity, combine $s_i = \sum_q w_q R_i^{(q)}$ with $\sum_q w_q = 1$ and $\sum_q w_q q^{-\beta} = 0$. A bounded-weight expansion $\mu(q) = \theta_\infty + cq^{-\beta} + O(q^{-2\beta})$ leaves $O(q_{\min}^{-2\beta})$ bias, which must be negligible relative to the standard error along with fitting error. Smooth uniform controls give $\beta = 2$; Gaussian quantile bins have order $1/(q \log q)$ instead. Both exponents and splines are prespecified screens, not selected by significance.

a) Complete audit procedure: Fix the information row in Table I, fitting rule and sampling bound before evaluation. Record copies and support, compute residual scores, and distinguish theorem-supported inputs from nominal screens. Set abstentions to $p^* = 1$ and apply BY to the complete family. Retain the raw values and validity status. A raw $p < \alpha$ is a *nominal positive*; retention uses BY.

TABLE III

FEEDBACK NULL: 400 INDEPENDENT ENTITIES, 20 OBSERVATIONS, FIVE FOLDS AND LOOKBACK FIVE. EQUAL ENTITY WEIGHTS AND IDENTICAL MIDDLE ROWS; 500 PAIRED REPLICATIONS.

Mean fitting	Exact mean	Mean T	Rejections
Complementary	-0.045052	-2.677	$0/500$
Gapped complementary	0.065533	3.148	$458/500$
Directional	0	-0.002	$27/500$

TABLE IV

MATCHED SAMPLING COMPARISON: EIGHT CATEGORIES, 30,976 OBSERVATIONS AND 300 PAIRED REPLICATIONS. BOTH METHODS USE THE SAME SIGNED LEARNING ALLOWANCE.

Sampling bound	Detections
Signed variance bound	$130/300$
Classical betting mixture	$296/300$

TABLE V

COMPLETE-FAMILY COMPARISON: 400 INDEPENDENT ENTITIES, 11 MODELS, 1,000 EVALUATION FAMILIES. FDR COLUMNS USE THE ALL-NULL LAW; POWER USES THREE EFFECTS OF 0.06. ALL RULES USE NOMINAL 0.05. CALIBRATION USES A SEPARATE 5,000-DRAW SIMULATOR BANK.

Fitting / moment	Normal FDR	Calibrated FDR	Calibrated power
Complementary	0.000	0.008	0.844
Gapped complementary	0.928	0.015	0.694
Directional blocks	0.011	0.009	0.666
Classical FOD	0.012	0.015	0.752

E. Multiple testing and selection

Proposition 4 (Complete-family error control)**.** *For fixed K and super-uniform true-null p-values, BY uses the step-up thresholds $j\alpha/(KH_K)$, $H_K = \sum_{j=1}^{K} 1/j$, and controls FDR at α under arbitrary dependence. Fixed K and asymptotically valid inputs give asymptotic control.*

All models remain in the family, including abstentions. Selecting the family or audit rule after viewing results requires a separate conditional-validity argument or independent confirmation.

IV. EXPERIMENTS

A. Which parts of the audit change the evidence?

Figure 2 contrasts resampled and fixed peers. For fixed peers, shared scores are centered; distinct scores depend on channel-effect alignment.

a) Reference reuse and false positives: For $Z \sim \mathrm{Bernoulli}(1/2)$ and conditionally independent $V, W \sim \mathrm{Bernoulli}(0.2+0.6Z)$, $\vartheta = 0$, $m_V = m_W = 0.35+0.3Z$ and $\Gamma = 0.0225$. With $M = 400, N = 64$, independent empirical-midrank training and 1,000 paired panels, shared/distinct rejections are $402/41$ with 8,192 training rows and $677/527$ with 32. All 48 configurations remain in the release.

For uniform $V \in \{0,1,2,3\}$, $W = (1,3,0,2)_V$ and constant controls, $\vartheta = 0$ despite dependence. Full and focal-only variances are $25/4096$ and $9/4096$. Across 3,000 panels with $M = 400, N = 64$, the full group test rejects 143 times, versus 482 with a focal-only scale.

b) Added value of signed learning allowances: A paired 2×2 ablation on previously inspected simulations crosses absolute/signed allowances with range/variance sampling bounds. Two balanced or eight unequal categories, five effects, 25–1,600 groups of 64 peers, 8,192 training rows, and 512 or 8,192 validation pairs give 100 configurations with estimated or shifted fits. Each construction uses the same observations and 1,000 repetitions. At 50,176 observations, two categories and $\vartheta = 0.01$, the four counts are 58, 78, 1,000 and 1,000: the large increase comes from sampling precision. At 30,976 observations and eight categories with $\vartheta = 0.013655$, both range counts are zero; absolute and signed variance bounds reject 274 and 435 times. The paired increase is 0.161 with pointwise 95% interval $[0.131, 0.189]$. This isolates learning-error handling at fixed sampling precision. Figure 3(a) shows the matched learning comparison; the companion retains the full component plots. All 160 nonpositive method cells have zero observed rejections, not a zero-risk guarantee.

c) Comparison with classical sampling methods: Classical alternatives remain competitive. At that eight-category budget, independent-triple and pooled-U variance counts are 433 and 461. A matched 300-replication extension fixes a 64-component bounded-mean betting mixture [17], with $p = \min(1, \delta + 1/E)$ after the same signed learning allowance.

d) Aggregate validation precision: Figure 3(b) shows 72 independently checked settings, 2,000 repetitions each and failure level $\delta = 0.05$. At 32 balanced categories, 8,192 validation pairs and exact fits, median bias slack falls from 0.018517 to 0.002602. Rectangles give smaller median slack in 10/72 settings, including all six two-category cases with opposite fitting errors. All 144 method cells have zero observed noncoverage; this does not assert zero risk.

e) Temporal feedback: With independent Gaussian innovations and $X_t = a_X + 0.2\sum_{h=1}^{5} Y_{t-h} + \xi_t$, removing one lookback from both sides still leaves a sign-reversing fitting-error product (Table III). Directional finite-sample null rates range from 0.054 to 0.088; the worst is 44/500 with Wilson 95% interval $[0.066, 0.116]$.

f) Complete-family error control: The raw-score branch evaluates BY on 10 or 11 forecasts, 25, 100 or 400 independent entities and 20 observations. Each cell uses 1,000 families and an independent 5,000-draw null bank. Forecast noises have correlation 0.5; three positive controls add τ times current innovation. Table V separates nominal and measured error. Supplied-null calibration requires the simulator law; reported Monte Carlo intervals condition on its realized bank. The classical comparator [24] averages $c_t(X_t - \bar{X}_{<t})(Y_t - \bar{Y}_{>t})$, where $c_t = \sqrt{(20-t)/(21-t)}$, over the same $t = 5, \ldots, 16$, normalized by $\sum_t c_t$. Its instrument is predictable under the null. It shares the constant-effect target but weights heterogeneous effects differently, and has higher power here.

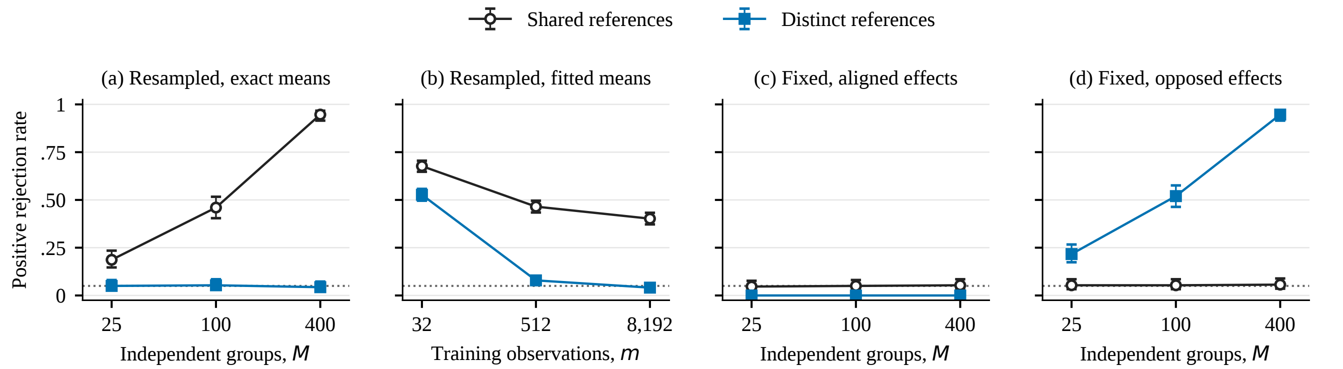

Fig. 2. **Reference correction depends on sampling and mean estimation.** Positive null rejection rates with pointwise Wilson 95% intervals; the dotted line is 0.05. Panels (a), (c), (d) use exact means, $N = 32$ and 300 replications. Panel (b) uses estimated means, $N = 64$, $M = 400$ and 1,000 replications at each training size. Shared and distinct references use the same evaluation panels. Fixed-peer effects are aligned or opposed between the two channels.

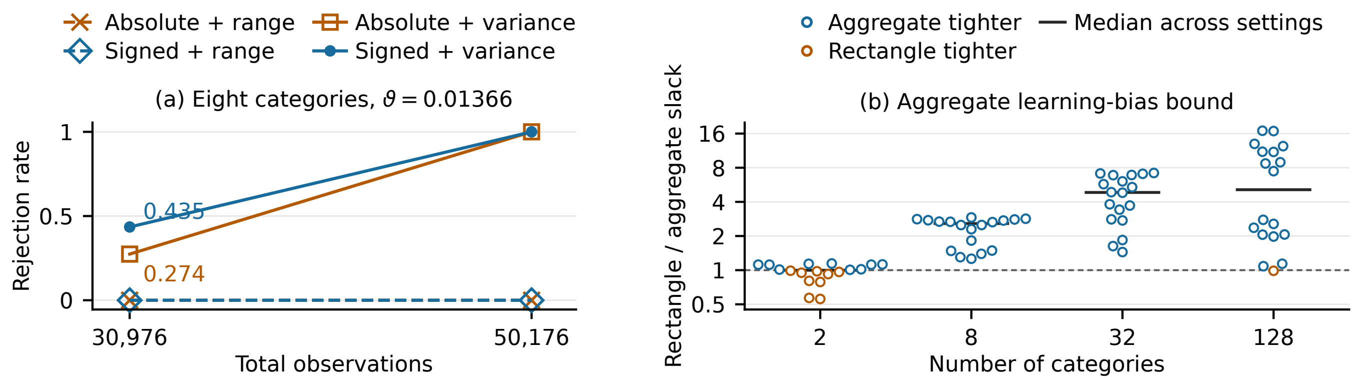

Fig. 3. Learning allowances and validation precision. (a) Eight-category factorial ablation, with 1,000 repetitions per point; lines connect tested budgets only. (b) All 72 matched validation settings, each with 2,000 repetitions. Slack is the upper bias bound minus true bias. Ratios above one favor aggregation; black marks are medians across 18 settings. Aggregation is tighter in 62 settings and rectangles in 10. Horizontal offsets separate points.

g) Does the adjustment capture the baseline?: An omitted entity–baseline interaction gives 300/300 nominal positives under every additive adjustment. In a near-copy rank null with 25 periods, 80 entities and noise 0.01, degrees-of-freedom correction and t_{24} give 982/1,000 rejections for 32 bins, 46 for linear-plus-spline adjustment and 48 for splines. The companion retains the full grid and scalar comparisons. Their different targets and sizes preclude a general calibration or power ranking.

B. Public forecast protocol

All 41 forecasts use five contiguous cluster folds, additive entity and baseline-bin controls, both exponents on $\{8, 12, 16, 24, 32\}$, and splines. Undefined ranks, negligible residuals, standard error $\leq 10^{-10}$ or observed copies receive $p^* = 1$. Directional comparisons fix bins and copy flags. Middle support means folds 2–4; empty training sides use pooled complementary bin means. The release records these fallback rows and unseen entities, which are outside the temporal centering result.

The four applications are UCI Electricity [29] (11 models, 48 meters, 34,992 next-day predictions), M5 [30] (10 models, 4,000 series, 112,000 weekly predictions), MovieLens-25M [31] (10 models, 12,088 ratings, 279 user clusters) and Open Source Asset Pricing [32] (10 models, 209 portfolios, 132 monthly origins). Their baselines are seven-day demand, last-week sales, training item mean and compounded past-year return. HAC lags are 14 daily, 2 weekly and 12 monthly; ratings use user clustering. The release specifies training cutoffs, provider versions and all forecast inputs.

V. Results and Analysis

A. Same forecasts, different audit decisions

Complementary-fold nominal screens retain identical sets under both exponents and splines. Directional sensitivities use $\beta = 2$ and the original scale; two-way-cluster checks use standard fits. Neither jointly validates both changes.

On all folds, complementary fitting retains 10/11, 7/10, 6/10 and 0/10 models in electricity, retail, ratings and portfolios, with 1, 2, 4 and 1 abstentions.

Figure 4 uses identical middle folds. Retail price/calendar changes from 6.41 to −0.82 and Croston SBA from −19.43 to 6.11; three models enter and one leaves the retained

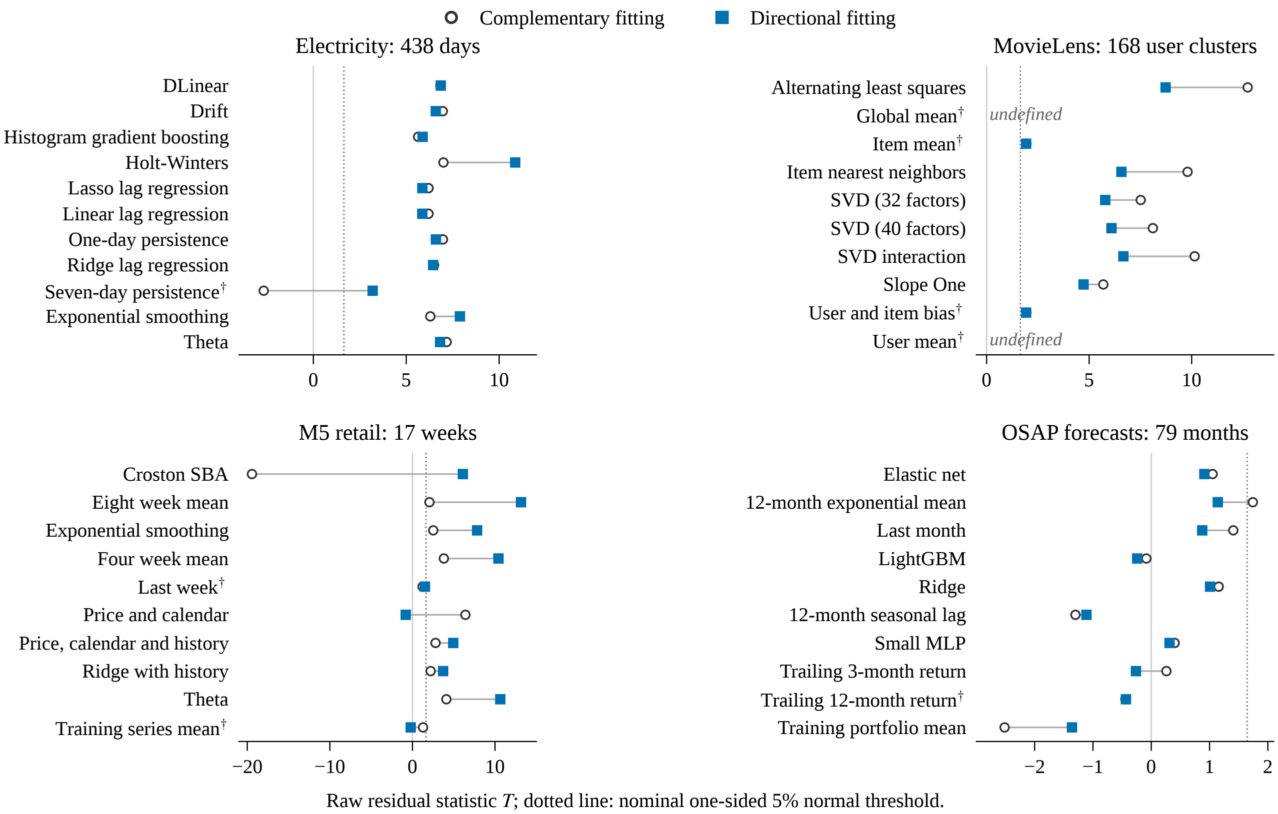

Fig. 4. **Same forecasts and observations, different nuisance training.** All 41 models use the middle three folds, identical ranks and bins, $\beta = 2$ and the original uncertainty rule. Lines join raw statistics; the dotted line is the nominal 5% normal threshold. Daggers mark fixed full-cohort abstentions; two pairs are undefined. MovieLens orders user clusters. These are sensitivity comparisons, not calibrated discoveries.

set. Their mean products change from 0.01014 to -0.00158 and from -0.01988 to 0.00552, respectively, on the same $68{,}000$ rows over 17 weeks. At each resolution, let r_X, r_Y be complementary residuals and Δ_X, Δ_Y the changes in fitted means. Then

$$\Delta\bar{R} = -\overline{r_X\Delta_Y} - \overline{r_Y\Delta_X} + \overline{\Delta_X\Delta_Y}. \tag{15}$$

The same extrapolation weights give $-0.00382, -0.00449, -0.00342$ for price/calendar and $0.01242, 0.00970, 0.00327$ for Croston. Each term reinforces the reversal. These are fit changes, not causal feedback identified by Proposition 3.

a) Accuracy and residual association: The ratings item-mean baseline has mean absolute error 0.693 and abstains; alternating least squares has error 0.719 and is retained under all four fold/support choices. Thus association and prediction accuracy differ. The electricity copy has raw $T = 28.72$ with ten-bin adjustment and 4.94 with splines before guarding. The same copy has $T = -24.85$ and -5.36 under the two exponents. These degenerate scores are abstained and do not calibrate nondegenerate nulls.

B. From residual evidence to forecast decisions

Independent replacement draws implement the reference design for a fixed archive, even when its original rows are dependent. In a $70{,}080$-row electricity archive, pooled U for boosting trained on $67{,}392$ earlier rows gives $\bar{D} = 0.078359$, $\widehat{B}_+ = 0.004277$ and $r_U = 0.003579$, hence $L_{0.05} = 0.07050$; the exact target is 0.07822. All seven bounds also retain the seasonal forecast: target 0.07741, pooled variance lower bound 0.06942. A category-only copy has target zero and lower bound -0.00794, and is not retained. The four categories cross weekend status with zero/positive seasonal forecasts; they do not represent the complete seasonal baseline. The $32{,}768$ queries access $26{,}140$ distinct rows of the already materialized archive; census takes 0.039 seconds versus 0.328 for the pooled boosting statistic.

Appliances energy [33] and Metro traffic [34] protocols were locally fixed before first download. A later integer-midrank repair corrected six copy decisions, preserving models and time windows; this was not third-party preregistration. Each study uses eight candidates, seasonal and Ridge baselines, past-hour features assumed available at hour end, and chronological training, calibration, selection and confirmation. The 32 categories cross 16 training baseline-forecast quantile bins with weekend status. Each gate uses $32{,}768$ replacement draws from the selection archive; BY applies to all eight candidates. Exact forecast metadata permit a classical covariance kernel [18] and pairs without learned means. They use all

TABLE VI
CONFIRMATION MSE AFTER TRAINING-SCALE CLIPPING; LOWER IS BETTER. PRIMARY RIDGE BASELINE, 1,008 ENERGY AND 8,713 TRAFFIC HOURS. GROUPED RULES DELIVER IDENTICAL FORECASTS IN BOTH TASKS.

Rule	Energy	Traffic
Reference U / betting	0.020804	0.005252
Known-forecast U / pairs	0.017277	0.001372
Direct-loss bound	0.020804	0.001372
Ungated augmentation	0.017277	0.001372
Convex combination	0.017277	0.001404
Ridge only	0.020804	0.005252

sampled rows for evaluation; reference methods split training, validation and evaluation.

Losses are scaled and clipped at the training 99th percentile. Table VI reports the two predeclared primary Ridge comparisons. Known-forecast U, known-forecast pairs and ungated selection all deliver the same ExtraTrees augmentation. It reduces observed confirmation MSE by 16.95% and 73.87% against Ridge, with zero additional gain from gating. The reference bounds revert to Ridge: for ExtraTrees their learning allowances 0.03810 and 0.03705 exceed observed associations 0.00995 and 0.00408. The predefined criterion of beating ungated and convex selection on both tasks is not met. Complete selection archives supply losses, so query counts do not establish label savings. Applying the joint rank-mean bound to all 32 candidates after observing these results reduces allowances by a median 1.24% (maximum 6.52%). All reference p-values remain one; this refinement changes no decision.

a) Development diagnosis with category metadata: After inspecting the tasks, we compare aggregate and rectangular allowances using identical exact category masses, fits, $32{,}768$ queries, evaluation scores and classical betting. Across four eight-candidate families, retentions rise from 3 to 13. For Appliances ExtraTrees beyond Ridge, the allowance falls from 0.029275 to 0.004328, with BY-adjusted betting $p = 0.003062$; Metro ExtraTrees remains unretained. This exposed-data diagnosis does not replace the original two-task criterion or establish new gating gains.

b) Rental-demand comparison: The earlier Seoul study [35] freezes 16 baseline/candidate pairs and a 2,016-hour confirmation window. Ridge augmentation gives 22,431 raw-count MSE gain and equals ungated selection. The seasonal candidate passes the copy guard, but its eight-week t_7 value 0.00668 becomes 0.1094 under BY, causing 299,790 MSE opportunity loss. For Ridge, gain over convex blending is 5,624; the weekly interval $[833, 10{,}414]$ excludes zero whereas the fortnight interval $[-898, 12{,}145]$ includes it. These temporal approximations and all strategies remain in the companion. Augmentation gain and gating gain are separate quantities.

c) Reproduction and cost: The release distinguishes source-data training from aggregate replay. Single-model audit time rises from 0.16 seconds near $31{,}000$ rows to 14.2 seconds at one million. Sorting matches exact pair calculations including ties; runtime settings accompany the $O(N \log N)$ bound.

VI. CONCLUSION

A residual audit must account for association created by evaluation itself. Our reference and feedback identities locate that association; aggregate validation and classical sampling bounds yield a computable reference certificate. Matched experiments and public forecasts separate the value of correcting an audit from the value of improving predictions.

a) Limitations: Observed-panel ranks, additive controls and fixed uncertainty choices lack joint calibration; their projection target can differ from full adjustment. Reference and directional guarantees need their stated sampling and training conditions, not just peer recurrence or time ordering. Rare categories, weak signals, dependence and near-copies remain difficult. Supplied-null calibration needs its external null law. Aggregate validation needs exact category masses and independent pairs; the joint refinement is not sharp. Archive certificates concern declared categories, without demonstrated acquisition savings. The original confirmations show no incremental gating gain; the new aggregate diagnosis uses already inspected data. Temporal loss bounds concern a same-period conditional average. Restricted training and historical-availability qualifications remain in the companion. Non-retention does not establish absence of predictive value.

b) Code and data availability: The fixed release provides code, licensed public data, all model and simulation results, and additional derivations.[1] All core statements and proofs are in this paper. Restricted insider-trading observations, identifiers, predictions, residuals and loaders remain private; only model-level and simulation aggregates are distributed.

APPENDIX
ASSUMPTIONS AND PROOFS

Proposition 4 applies the BY theorem [36], conditionally on selection when required, then averages. Marginal validity alone is insufficient: for uniform p and $\alpha < t$, $\Pr(p \le \alpha \mid p \le t) = \alpha/t$.

a) Finite rank references:

Proof of Proposition 1. For distinct i, j, l, conditional reference independence gives $\mathbb{E}[a_{ij} b_{il} \mid V_i, W_i, Z_i, H] = F_V(V_i) F_W(W_i)$. The score D_i averages $(a_{ij} - f(Z_i))(b_{il} - g(Z_i))$ over these ordered pairs. Conditional centering gives $\vartheta + b_H$. The k shared pairs among k^2 pairs in C_i add Γ/k.

Proof of Theorem 1. Symmetrize $(a_{12} - m_V(Z_1))(b_{13} - m_W(Z_1))$ over the six permutations. Conditional expectations for the focal and two reference roles are $r_V r_W, h_V(V), h_W(W)$, each with mean ϑ. The centered first projection is $\psi/3$. Bounded U-statistic decomposition [37] gives $D_j - \vartheta = N^{-1} \sum_i \psi(O_{ji}) + R_j$, with $\mathbb{E} R_j = 0$ and $\mathbb{E} R_j^2 = O(N^{-2})$. The scaled mean remainder has

[1] https://github.com/MaxwellNi/forecast-audit/releases/tag/icdm2026-v7.

variance $O(N^{-1})$; bounded first projections obey the CLT. Direct algebra gives $C_j - D_j = (J_j - Q_j)/(N-1)$, where J_j, Q_j average the shared comparison pair and distinct comparison triple, respectively. Their difference has mean Γ and variance $O(N^{-1})$. Its centered scaled mean has variance $O(N^{-2})$, proving both centered limits and $N\,\mathrm{Var}(A_j) \to \sigma_\psi^2$. Symmetrization into disjoint triples bounds centered fourth moments by $O(N^{-2})$. With $W_j = A_j - \mathbb{E}A_j$, $\mathrm{Var}[(N/M)\sum_j W_j^2] = O(M^{-1})$ and $N\bar{W}^2 = O_p(M^{-1})$. Hence $N\widehat{v}_A \to_p \sigma_\psi^2$. Slutsky gives the boundary limits; for the divergent case, $T_C/[\Gamma\sqrt{MN}/\{(N-1)\sigma_\psi\}] \to_p 1$. Conditional independence centers each reference-role residual. Along the stated arrays, boundedness makes all remainder and moment bounds uniform; the positive limiting variance gives Lindeberg's condition.

Proof of Corollary 1. Let $\delta_f = f - m_V$, $\delta_g = g - m_W$, and let Δ_j be the fitted-minus-oracle group score. This difference is the same for C, D:

$$\Delta_j = \frac{1}{N}\sum_i \{ -\delta_f(Z_i)(U_{W,i} - m_W(Z_i)) - \delta_g(Z_i)(U_{V,i} - m_V(Z_i)) + \delta_f(Z_i)\delta_g(Z_i)\}.$$

It is an order-two U-statistic with conditional mean b_H. Bounded comparisons and clipped fits give $\mathrm{Var}(\Delta_j \mid H) \le C(\varepsilon_f^2 + \varepsilon_g^2)/N$ for an absolute constant C. Conditioning on the common independent training sample preserves group independence, so $\sqrt{MN}(\bar{\Delta} - b_H) = o_p(1)$. The product-rate condition gives $\sqrt{MN} b_H = o_p(1)$. For the sample variance $\widehat{v}_\Delta$, conditional unbiasedness gives $N\widehat{v}_\Delta = o_p(1)$. Sample covariance Cauchy–Schwarz bounds $N|\widehat{v}_A - \widehat{v}_A^o|$ by $N\widehat{v}_\Delta + 2\sqrt{N\widehat{v}_\Delta\, N\widehat{v}_A^o} = o_p(1)$. The oracle limits and ratio argument therefore transfer. For fixed finite controls with positive masses and training size m, each empirical-midrank cell mean is a ratio of bounded order-two U-statistics and cell proportions. Its root-m rate gives the product condition when $m/\sqrt{MN} \to \infty$.

Proof of Proposition 2. For the rectangular choice, conditional on focal validation categories, pair comparisons have means $m_V(c), m_W(c)$. Their intervals fail with total probability at most $\delta/2$. The binomial limits fail with probability at most $\delta/2$. The limits are $u_c = \mathrm{Beta}^{-1}(1 - \delta/(2C_Z); n_c + 1, m_{\mathrm{val}} - n_c)$ for $n_c < m_{\mathrm{val}}$, and one otherwise. They satisfy $u_c \ge n_c/m_{\mathrm{val}}$, so $\mathcal{P}$ is feasible. On the joint event the true probability vector lies in $\mathcal{P}$; bilinearity gives the endpoint extrema and $\widehat{B}_- \le b_H \le \widehat{B}_+ \le \widehat{B}_{\mathrm{abs}}$.

Condition on independent training and validation, denoted H. Each disjoint-triple kernel has variance σ^2 and range at most R. Permutation averaging represents $\bar{D}$ as an average of means of J independent kernels. Exponential convexity transfers their Hoeffding and Bernstein moment bounds to $\bar{D}$ [38]. Consequently, its upper deviation exceeds either $R\sqrt{x/(2J)}$ or $\sigma\sqrt{2x/J} + Rx/(3J)$ with probability at most e^{-x}. Theorem 10 of [15] gives $\sigma \le s + R\sqrt{2x/(J-1)}$ except with probability e^{-x}. For fixed H, compare the Hoeffding radius with the Bernstein radius using the true σ. Apply the smaller fixed-radius tail, and then the variance event. Their union yields the minimum in Eq. (11) with failure at most $2e^{-x}$; independence between s and $\bar{D}$ is unnecessary. Since $\mathbb{E}[\bar{D} \mid H] = \vartheta + b_H$, adding δ proves the bound.

For explicit inversion, set $d = (\bar{D} - \widehat{B})_+$, $A = s\sqrt{2/J}$, $x_H = 2J(d/R)^2$ and $x_B = \{2d/(A + \sqrt{A^2 + 4c_J d})\}^2$. For $d > 0$, $p_{\mathrm{cert}} = \min\{1, \delta + 2e^{-\max(x_H, x_B)}\}$; set it to one for $d = 0$. Monotone inversion gives super-uniformity. The range-only version uses $\min\{1, \delta + e^{-x_H}\}$.

Proof of Lemma 1. If $\mathcal{S}$ is empty, the deterministic corner bound suffices. Otherwise, condition only on the focal validation categories, and write $a_c = m_V(c) - f(c)$, $b_c = m_W(c) - g(c)$ and $e_{Ac} = \widehat{a}_c - a_c$, $e_{Bc} = \widehat{b}_c - b_c$. The errors are independent across streams and categories, with sub-Gaussian proxies $1/(4n_{Ac})$ and $1/(4n_{Bc})$ by Hoeffding's lemma. Over $\mathcal{S}$,

$$\sum_c p_c(a_c b_c - \widehat{a}_c\widehat{b}_c) = -\sum_c p_c\widehat{b}_c e_{Ac} - \sum_c p_c\widehat{a}_c e_{Bc} + \sum_c p_c e_{Ac} e_{Bc}.$$

Conditioning on the opposite stream gives upper-tail bounds r_A, r_B, each failing with probability at most e^{-x}. For independent centered sub-Gaussian X, Y with proxies s_X^2, s_Y^2, condition on X and integrate a standard normal variable to obtain $\mathbb{E}e^{tXY} \le (1 - t^2 s_X^2 s_Y^2)^{-1/2}$ for $|t| s_X s_Y < 1$. Thus $R_2 = \sum_c p_c e_{Ac} e_{Bc}$ obeys

$$\log \mathbb{E}e^{tR_2} \le \frac{t^2 \sum_c d_c^2}{2(1 - t\max_c d_c)}, \qquad 0 < t < 1/\max_c d_c.$$

Here $-\log(1 - u^2) \le u^2/(1 - |u|)$. Chernoff optimization gives $\Pr(R_2 > r_2) \le e^{-x}$. A union bound over the three tails gives $3e^{-x} = \delta$ without requiring their events to be independent. Missing categories contribute at most their deterministic corner bound; conditioning fixes the missing set. Remove conditioning to finish. For Eq. (10), exact fits give centered stream means with second moments at most $C/(4n)$. Jensen's inequality bounds each expected linear radius by $\sqrt{Cx/8}/n$; the product radius equals $\sqrt{Cx/8}/n + x/(4n)$. Adding gives the result.

b) Temporal splitting:

Proof of Proposition 3. Since $y_r - \bar{y}_O = \varepsilon_r - \bar{\varepsilon}_O$, the entity effect cancels from the outcome residual. Its covariance with the forecast residual depends only on the shocks, because the remaining terms are uncorrelated with them. For each $r \in I$, expansion gives

$$\frac{\mathbb{E}[(x_r - \bar{x}_O)(y_r - \bar{y}_O)]}{\sigma^2} = -\frac{\sum_{j\in O} w_{rj}}{m} - \frac{\sum_{s\in O} w_{sr}}{m} + \frac{\sum_{s,j\in O} w_{sj}}{m^2}.$$

The first sum uses only E and the second only L, proving Eq. (14) after summing over I. With temporal splitting,

$x_r - \bar{x}_E$ uses only shocks before r, whereas $y_r - \bar{y}_L$ uses shocks at or after r; orthogonality gives zero expectation. For the nonlinear case, let $\mathcal{G}_s$ be the conditioning field in the proposition. The forecast residual is $\mathcal{G}_s$-measurable for every $s \geq r$, so $\mathbb{E}[(x_r - \bar{x}_E)\varepsilon_s] = 0$ by iterated expectation. Apply this at $s = r$ and average over $s \in L$. For the closed form, let $S = a\{Khp - p(p+1)/2\}$ be the sum of all forecast weights and $B = a(K-1)p(p+1)/2$ the sum crossing fold boundaries. Summing Eq. (14) yields $\sigma^2\{(K-1)S - (2K-1)B\}/\{(K-1)^2 h\}$, which simplifies to the stated value.

Proof of Corollary 2. Each S_g has mean zero. The bounded fourth moments make the Lyapunov ratio $O(G^{-1})$. Independence also gives $G^{-1}\sum_g\{S_g^2 - \mathbb{E}S_g^2\} \to_p 0$ and $\bar{S} \to_p 0$. Thus the sample variance converges to v; studentization gives the limit.

c) Fixed peers: For the fixed-peer statement, condition on the entity attributes. Channel independence implies $\mathbb{E}[U_{V,i}U_{W,i}] = \bar{p}_i\bar{q}_i$ and $\mathbb{E}[a_{ij}b_{ij}] = p_{ij}q_{ij}$, where $\bar{p}_i = k^{-1}\sum_{j\neq i} p_{ij}$ and similarly for $\bar{q}_i$. Thus exact entity centering gives $\mathbb{E}C_i = 0$; substituting into Eq. (3) proves Eq. (13).

References

[1] R. C. Fair and R. J. Shiller, "Comparing information in forecasts from econometric models," *American Economic Review*, vol. 80, no. 3, pp. 375–389, 1990.

[2] R. D. Shah and J. Peters, "The hardness of conditional independence testing and the generalised covariance measure," *The Annals of Statistics*, vol. 48, no. 3, pp. 1514–1538, 2020.

[3] R. Giacomini and H. White, "Tests of conditional predictive ability," *Econometrica*, vol. 74, no. 6, pp. 1545–1578, 2006.

[4] V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins, "Double/debiased machine learning for treatment and structural parameters," *The Econometrics Journal*, vol. 21, no. 1, pp. C1–C68, 2018.

[5] H. D. Chiang, K. Kato, Y. Ma, and Y. Sasaki, "Multiway cluster robust double/debiased machine learning," *Journal of Business & Economic Statistics*, vol. 40, no. 3, pp. 1046–1056, 2022.

[6] K. Zhang, J. Peters, D. Janzing, and B. Schölkopf, "Kernel-based conditional independence test and application in causal discovery," in *Uncertainty in Artificial Intelligence*, 2011, pp. 804–813.

[7] L. Petersen and N. R. Hansen, "Testing conditional independence via quantile regression based partial copulas," *Journal of Machine Learning Research*, vol. 22, no. 70, pp. 1–47, 2021.

[8] A. R. Lundborg, I. Kim, R. D. Shah, and R. J. Samworth, "The projected covariance measure for assumption-lean variable significance testing," *The Annals of Statistics*, vol. 52, no. 6, pp. 2851–2878, 2024.

[9] C. Scheidegger, J. Hörrmann, and P. Bühlmann, "The weighted generalised covariance measure," *Journal of Machine Learning Research*, vol. 23, no. 273, pp. 1–68, 2022.

[10] Z. He, R. Pogodin, Y. Li, N. Deka, A. Gretton, and D. J. Sutherland, "On the hardness of conditional independence testing in practice," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.

[11] M. Pan, A. de Mathelin, and W. Tansey, "Empirically calibrated conditional independence tests," in *International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 300, 2026, pp. 5194–5202.

[12] W. R. Schucany, H. L. Gray, and D. B. Owen, "On bias reduction in estimation," *Journal of the American Statistical Association*, vol. 66, no. 335, pp. 524–533, 1971.

[13] X. Ou and R. Nabi, "Coarsening bias from variable discretization in causal functionals," in *Proceedings of the Forty-Second Conference on Uncertainty in Artificial Intelligence*, 2026.

[14] I. Kim, M. Neykov, S. Balakrishnan, and L. Wasserman, "Local permutation tests for conditional independence," *The Annals of Statistics*, vol. 50, no. 6, pp. 3388–3414, 2022.

[15] A. Maurer and M. Pontil, "Empirical Bernstein bounds and sample variance penalization," in *Proceedings of the 22nd Annual Conference on Learning Theory*, 2009. [Online]. Available: https://www.cs.mcgill.ca/~colt2009/papers/012.pdf

[16] T. Peel, S. Anthoine, and L. Ralaivola, "Empirical Bernstein inequalities for U-statistics," in *Advances in Neural Information Processing Systems*, vol. 23, 2010. [Online]. Available: https://papers.neurips.cc/paper_files/paper/2010/hash/d6ef5f7fa914c19931a55bb262ec879c-Abstract.html

[17] I. Waudby-Smith and A. Ramdas, "Estimating means of bounded random variables by betting," *Journal of the Royal Statistical Society Series B*, vol. 86, no. 1, pp. 1–27, 2024.

[18] E. Schechtman and S. Yitzhaki, "A measure of association based on Gini's mean difference," *Communications in Statistics: Theory and Methods*, vol. 16, no. 1, pp. 207–231, 1987.

[19] I. Kim, L. Wasserman, S. Balakrishnan, and M. Neykov, "Semi-supervised U-statistics," *The Annals of Statistics*, vol. 53, no. 6, pp. 2488–2515, 2025.

[20] L. Liu, R. Mukherjee, and J. M. Robins, "Assumption-lean falsification tests of rate double-robustness of double-machine-learning estimators," *Journal of Econometrics*, vol. 240, no. 2, p. 105500, 2024.

[21] J. Robins, L. Li, R. Mukherjee, E. Tchetgen Tchetgen, and A. van der Vaart, "Higher order estimating equations for high-dimensional models," *The Annals of Statistics*, vol. 45, no. 5, pp. 1951–1987, 2017.

[22] S. Nickell, "Biases in dynamic models with fixed effects," *Econometrica*, vol. 49, no. 6, pp. 1417–1426, 1981.

[23] B. S. So and D. W. Shin, "Recursive mean adjustment in time-series inferences," *Statistics & Probability Letters*, vol. 43, no. 1, pp. 65–73, 1999.

[24] M. Arellano and O. Bover, "Another look at the instrumental variable estimation of error-components models," *Journal of Econometrics*, vol. 68, no. 1, pp. 29–51, 1995.

[25] V. Semenova, M. Goldman, V. Chernozhukov, and M. Taddy, "Inference on heterogeneous treatment effects in high-dimensional dynamic panels under weak dependence," *Quantitative Economics*, vol. 14, no. 2, pp. 471–510, 2023.

[26] D. Leung and M. Drton, "Testing independence in high dimensions with sums of rank correlations," *The Annals of Statistics*, vol. 46, no. 1, pp. 280–307, 2018.

[27] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[28] W. K. Newey and K. D. West, "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix," *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.

[29] A. Trindade, "ElectricityLoadDiagrams20112014," UCI Machine Learning Repository, 2015, doi: 10.24432/C58C86.

[30] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "M5 accuracy competition: Results, findings, and conclusions," *International Journal of Forecasting*, vol. 38, no. 4, pp. 1346–1364, 2022.

[31] F. M. Harper and J. A. Konstan, "The MovieLens datasets: History and context," *ACM Transactions on Interactive Intelligent Systems*, vol. 5, no. 4, pp. 19:1–19:19, 2015.

[32] A. Y. Chen and T. Zimmermann, "Open source cross-sectional asset pricing," *Critical Finance Review*, vol. 11, no. 2, pp. 207–264, 2022.

[33] L. Candanedo, "Appliances energy prediction," UCI Machine Learning Repository, 2017.

[34] J. Hogue, "Metro interstate traffic volume," UCI Machine Learning Repository, 2019.

[35] "Seoul Bike Sharing Demand," UCI Machine Learning Repository, 2020.

[36] Y. Benjamini and D. Yekutieli, "The control of the false discovery rate in multiple testing under dependency," *The Annals of Statistics*, vol. 29, no. 4, pp. 1165–1188, 2001.

[37] W. Hoeffding, "A class of statistics with asymptotically normal distribution," *The Annals of Mathematical Statistics*, vol. 19, no. 3, pp. 293–325, 1948.

[38] ——, "Probability inequalities for sums of bounded random variables," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 13–30, 1963.